

App, we will perform some network operations to pull the feeds from the Internet. As soon as the main activity loads a request would be made to the server and response will be awaited. Now, because Android is based on a single thread model, our screen would basically freeze till the time response is awaited. Hence, performing long running operations on the UI thread must be avoided. This also takes into account file and network access.

To get over such situation, you will have to create a new thread and implement run method to perform such network operations, so that the UI stays responsive. But the Android UI toolkit is not thread safe, so if we ever need to make some change to the UI on the basis of the result of the background operation performed, then this approach will lead to some serious issues. To resolve these issues very good pattern which is enveloped into **AsyncTask**. An **AsyncTask** is defined by a computational code that runs on some background thread and whose result can be easily published on the UI thread.

AsyncTask generally has four basic steps:

1. doInBackground: The main part of the code which is to perform long running operation is declared in this method. The **onClick** method when executed, calls **execute** method which accepts parameters and calls the **doInBackground** method automatically, with the parameters passed.

2. onPostExecute: After the **doInBackground** method has completed processing, this method is being called. The result from the **doInBackground** is passed to the **onPostExecute** method.

3. onPreExecute: This method is called must before the **doInBackground** method is invoked.

4. onProgressUpdate: This method can be invoked by calling **publishProgress**, whenever required, from **doInBackground** call this method.

XmlPullParser

XML stands for Extensible Markup Language and is a reputed data exchange format. An XML document typically consists of elements - every element has a start tag, content and an end tag. An XML document will have exactly one root element and it is case-sensitive. An XML file is considered valid if